

BRIDGING THE GAP IN S&OP 4.0: A DECISION INTELLIGENCE ARTIFACT FOR ADVANCED FORECASTING AND SCENARIO SIMULATION

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ABSTRACT

Sales and Operations Planning (S&OP) is essential for building resilient supply chains, yet traditional forecasting systems often fail to align mathematical accuracy with actual business risk. This Work-in-Progress paper adopts the Design Science Research (DSR) methodology to propose an on-premises Decision Intelligence artifact designed for S&OP 4.0. The system introduces an automated model tournament that evaluates various machine learning and econometric algorithms. However, instead of optimizing solely symmetric error, the artifact translates metrics into an actionable multi-objective decision score. This innovation prevents algorithms from masking systematic directional tendencies that lead to critical stockouts or excessive working capital. Empirical evaluation was conducted using open data from ANFAVEA (National Association of Motor Vehicle Manufacturers) covering Brazilian passenger cars and light commercial vehicles disaggregated into fuel categories from January 2023 to August 2026. To strictly prevent data leakage and look-ahead bias, the dataset was partitioned using a temporal holdout strategy. Data from January 2023 to August 2025 was exclusively used for model training and parameter estimation, establishing the forecasting origin. The remaining 12-month period (September 2025 to August 2026) served as the unseen out-of-time test horizon, comprising a total actual validation volume of 2,864,628 vehicles. By optimizing via the asymmetric decision score, the artifact reduced the absolute physical misallocation by 34.1% sacrificing a negligible 0.15 percentage points in symmetric WMAPE while avoiding critical inventory imbalances.

KEYWORDS

S&OP 4.0, Decision Intelligence, Advanced Forecasting, Design Science Research, Machine Learning

1. INTRODUCTION

The global supply chain is characterized by dynamic complexity, requiring Sales and Operations Planning (S&OP) processes to evolve from static formal schedules into resilience-building mechanisms capable of adapting to changing conditions (Jonsson et al., 2021). While digitalization and Artificial Intelligence (AI) act as key project enablers for agile decision-making (Ouamalich & El Hachemi, 2025), a comprehensive synthesis of the literature reveals a persistent lack of unifying frameworks for S&OP maturity and performance measurement (Thomé et al., 2012).

Furthermore, S&OP encompasses two distinct facets: the "hard side," focused on data, schedules, and tools, and the "soft side," characterized by culture, leadership, and human collaboration (Tuomikangas & Kaipia, 2014). The implementation of advanced S&OP is frequently hindered by these soft issues, particularly concerning how decision-makers interact with complex data (Benaviba et al., 2025). The reliance on purely mathematical models without proper executive translation exacerbates this friction.

To address these gaps, this research proposes the development of a Decision Intelligence artifact. The system introduces an automated model tournament that generates mathematical scenarios while respecting the distinct behavioral properties of time series. By converting forecasting errors into a business-oriented decision score, the artifact aims to reduce cognitive friction, optimize input costs, and empower executive consensus.

2. RELATED WORK

Recent studies emphasize the urgency of upgrading tactical planning. Kalla et al. (2026) advocate for an "S&OP 4.0," leveraging Industry 4.0 technologies—such as machine learning and generative AI—to enable faster and more precise decision-making in complex networks. This technological shift is crucial for integrating supply chain risk management (SCRM) to effectively handle uncertain environments (Kalla et al., 2025). Additionally, Cabello et al. (2026) emphasizes the importance of scenario planning and sensitivities to prioritize responsive "fast-track" capacity, mitigating demand volatility and lead-time-driven stockouts.

The practical value of integrating robust quantitative methods has been proven empirically. Kačmárý & Kleinová (2026) demonstrated that implementing a forecasting-based planning system using ARIMA and qualitative expert corrections resulted in a 20% reduction in extracted stock within a magnesite mining operation. However, the effectiveness of these technical solutions relies heavily on cross-functional coordination (Bhalla et al., 2022) and contextualized information sharing, which economizes attention and improves overall responsiveness (Kaipia et al., 2017).

Despite these advancements, an effective bridge between academic and industry practice is needed. Kreuter et al. (2022) explicitly recommend the Design Science approach to integrate practitioners into academic research, enhancing the quality and relevance of evidence in S&OP studies.

3. RESEARCH METHODOLOGY

Addressing the call by Kreuter et al. (2022) to bring academic research and practice closer together, this study adopts the Design Science Research (DSR) methodology. DSR is an artifact-oriented discipline that aims to solve known problems or design novel operational systems. A core strength of DSR is its focus on eliminating the distance between theory and practice, intentionally generating prescriptive knowledge for specific organizational challenges (Goecks et al., 2021).

The proposed solution is classified as an Algorithm/Instantiation artifact, developed as a high-performance, on-premises advanced forecasting module built with Python and Streamlit to ensure absolute data sovereignty without recurring SaaS fees. The artifact's architecture follows a structured construction heuristic divided into four sequential phases: (1) Signal ingestion and transformation, (2) Model tournament, (3) Loss evaluation and decision score, and (4) Retraining and what-If simulation.

3.1 Phase 1: Signal Ingestion and Transformation

The artifact targets monthly univariate time series representing aggregate vehicle physical sales volume per product category (positive continuous real numbers $y_t \in \mathbb{R}^+$), while accommodating correlated exogenous macroeconomic regressors where applicable. Traditional forecasting often enforces a single universal pre-processing method across entire portfolios, sub-optimizing heterogeneous time series. To prevent this, the artifact introduces a combinatorial operator that evaluates eight concurrent pre-processing scenarios ($\mathcal{C} = \{C_1, \dots, C_8\}$) generated through function composition across three fundamental mathematical operations:

1. **Winsorization (W):** Mitigates extreme outliers based on the Interquartile Range ($IQR = Q_3 - Q_1$), where Q_1 and Q_3 denote the first (25th) and third (75th) quartiles, respectively. Applying a scaling factor $k = 1.5$, anomalous peaks and troughs are bounded without discarding sequence continuity:

$$y_t^{(wins)} = \max(Q_1 - k \cdot IQR, \min(y_t, Q_3 + k \cdot IQR))$$

2. **Logarithmic Transformation (L):** Stabilizes heteroscedastic variance and linearizes exponential trends:

$$y_t^{(log)} = \ln(y_t + \delta), \quad \delta > 0$$

3. **Normalization (N):** Maps values via Min-Max scaling into the closed interval $[0,1]$ to ensure numerical stability and convergence for scale-sensitive estimators:

$$y_t^{(norm)} = \frac{y_t - \min(y)}{\max(y) - \min(y)}$$

The eight deterministic transformation pipelines $C \in \mathcal{C}$ are formalized in Table 1.

Table 1. The Combinatorial Pre-Processing Scenarios

Scenario ID	Mathematical Composition	Sequential Pipeline	Strategic Purpose
C1 (Base)	y_t	Raw series	Preserves original signal without intervention.
C ₂ (W)	$W(y_t)$	$W(y_t)$	Mitigates peak/trough outlier distortion.
C ₃ (L)	$L(y_t)$	$L(y_t)$	Stabilizes exponential variance.
C ₄ (N)	$N(y_t)$	$N(y_t)$	Normalizes scale for neural/gradient models.
C ₅ (L + N)	$N(L(y_t))$	$L \rightarrow N$	Combines variance stabilization with uniform scale.
C ₆ (W + L)	$L(W(y_t))$	$W \rightarrow L$	Treats outliers before dampening variance.
C ₇ (W + N)	$N(W(y_t))$	$W \rightarrow N$	Truncates anomalies before bounded scaling.
C ₈ (W + L + N)	$N(L(W(y_t)))$	$W \rightarrow L \rightarrow N$	Full pipeline for highly noisy/volatile series.

Note on Pipeline Order: When multiple transformations occur, their execution sequence is strict: $W \rightarrow L \rightarrow N$. Outliers must be bounded (W) prior to non-linear dampening (L) to prevent extreme leverage, while scaling (N) must always be the final step to keep inputs strictly within $[0,1]$ for algorithmic execution.

3.2 Phase 2: The Model Tournament (S&OP 4.0 Engine)

In Phase 2, each time series under all eight transformation scenarios ($C \in \mathcal{C}$) is evaluated across an algorithm library $\mathcal{M}$ comprising 12 distinct estimators. This library combines classical econometric and benchmark techniques (AutoARIMA, TBATS, Theta, Exponential Smoothing, Naïve Mean, Naïve Seasonal, Naïve Last Value) with supervised machine learning algorithms (Prophet, XGBoost, LightGBM, Random Forest, Gradient Boosting).

This structure yields a combinatorial tournament matrix of $8 \times 12 = 96$ execution pipelines per product category. The models compete using an out-of-time temporal rolling-origin split to identify the optimal pair of data transformation and forecasting algorithm. When evaluating multivariate-capable algorithms (e.g., Prophet), the engine dynamically assesses feature correlation to ingest exogenous regressors X_t , generating independent forecasts $\hat{y}_t^{(m)}$ across the specified planning horizon h .

3.3 Phase 3: Loss Evaluation and The Decision Score

The primary innovation lies in translating mathematical loss into an actionable managerial metric. While standard data science pipelines optimize purely symmetric errors such as WMAPE, tactical S&OP demands forecasts free from chronic directional bias. The artifact converts accuracy and bias into a unified decision score, functioning as a multi-objective decision-making function (Ouamalich & El Hachemi, 2025).

First, symmetric accuracy is evaluated via Weighted Mean Absolute Percentage Error (WMAPE) for each model m over test periods $t = 1, \dots, N$:

$$\text{WMAPE}^{(m)} = \frac{\sum_{t=1}^N |y_t - \hat{y}_t^{(m)}|}{\sum_{t=1}^N |y_t|} \times 100$$

Simultaneously, directional percentage bias is extracted to capture systematic deviations across actual volume $V_{real} = \sum_{t=1}^N y_t$ and predicted volume $V_{pred} = \sum_{t=1}^N \hat{y}_t^{(m)}$:

$$\text{Bias}_{\%}^{(m)} = \left(\frac{V_{pred} - V_{real}}{V_{real}} \right) \times 100$$

Finally, the artifact evaluates models using an Asymmetric Decision Score (DS). In S&OP environments, the operational consequences of under-forecasting (leading to stockouts and lost sales) are fundamentally different from over-forecasting (leading to excess inventory and holding costs). To capture this economic reality, the artifact introduces an asymmetric directional penalty function:

$$\text{Penalty_Bias}_m = \begin{cases} C_u \times |\text{Bias}_{\%m}| & \text{if } \text{Bias}_{\%m} < 0 \text{ (Under-forecast)} \\ C_o \times |\text{Bias}_{\%m}| & \text{if } \text{Bias}_{\%m} \geq 0 \text{ (Over-forecast)} \end{cases}$$

Where the cost weights are calibrated based on business risk (e.g., $C_u > C_o$). For this preliminary validation, C_u was set to 1.5 to heavily penalize stockout risks, while C_o was set to 1.0. The champion model m^* minimizes this integrated objective function:

$$DS_m = WMAPE_m + Penalty_Bias_m$$

$$m^* = \operatorname{argmin}_{m \in M} DS_m$$

While multi-objective optimization frequently employs Pareto frontiers, generating a non-dominated continuous front introduces prohibitive cognitive friction and manual trade-off negotiations during time-constrained executive consensus meetings. The scalar decision score provides an unambiguous, automated ranking mechanism that directly mitigates organizational soft issues (Benaviba et al., 2025).

3.4 Phase 4: Retraining and What-If Simulation

Once the champion model m^* is determined, it is retrained on the full historical time series to generate the unconstrained statistical baseline $\hat{y}_{t+h}^{(base)}$. To incorporate commercial judgment, the user interface features a what-If simulation parameter (α). Rather than acting as an arbitrary scaling factor, α enables cross-functional teams to simulate specific market hypotheses (e.g., promotional campaigns, supply bottlenecks, commercial adjustments) on top of the statistical baseline, achieving collaborative human-in-the-loop S&OP consensus:

$$\hat{y}_{t+h}^{(S\&OP)} = \hat{y}_{t+h}^{(base)} \times \left(1 + \frac{\alpha}{100}\right)$$

4. PRELIMINARY RESULTS AND DISCUSSION

Empirical evaluation was conducted using open data from ANFAVEA (National Association of Motor Vehicle Manufacturers) covering Brazilian passenger cars and light commercial vehicles disaggregated into fuel categories from January 2023 to August 2026. To strictly prevent data leakage and look-ahead bias, the dataset was partitioned using a temporal holdout strategy. Data from January 2023 to August 2025 was exclusively used for model training and parameter estimation, establishing the forecasting origin. The remaining 12-month period (September 2025 to August 2026) served as the unseen out-of-time test horizon, comprising a total actual validation volume of 2,864,628 vehicles.

The automated tournament executed eight distinct scenario pipelines across the seven categories in 8 minutes and 37 seconds on a standard commercial laptop (Intel Core i7, 8 GB RAM), demonstrating that advanced analytics do not require costly high-performance computing infrastructure to operate effectively in tactical planning routines.

To establish a robust independent benchmark, the proposed Decision Intelligence artifact was compared against the current industrial standard approach: a pure symmetric optimization model. In this baseline (Standard WMAPE Optimization), models such as traditional ARIMA and Exponential Smoothing are selected solely based on their WMAPE performance, ignoring asymmetric operational costs. Table 2 details the comparative performance between this standard industry baseline and the models selected via the proposed multi-objective decision score.

Table 2. Comparison of forecast performance: WMAPE optimization vs. decision score (DS)

Victory Criterion	Consolidated WMAPE (%)	Test Volume	Net Bias	Absolute Bias	Bias %
WMAPE Optimization	8.06	2,864,628	-27,573	36,587	-0.96
Decision Score (DS)	8.21	2,864,628	-12,309	24,115	-0.43
Variation (Delta)	+0.15 p.p.	-	+15,264	-12,472 (-34.1%)	+0.53 p.p.

Optimizing strictly for WMAPE yielded an error of 8.06% but generated a high Absolute Bias of 36,587 misallocated vehicles across individual product families. Conversely, adopting the asymmetric decision score

accepted a minor increase of 0.15 percentage points in WMAPE (reaching 8.21%) while eliminating 12,472 units of physical misallocation—a 34.1% reduction in absolute bias. This trade-off significantly curtails systemic under-forecasting that triggers line stockouts and over-forecasting that ties up working capital, directly resolving the operational challenges raised by Cabello et al. (2026) and Kalla et al. (2025).

5. CONCLUSION AND FUTURE WORK

This study developed and empirically evaluated an on-premises Decision Intelligence artifact for S&OP 4.0. The results demonstrate that multi-objective evaluation combining symmetric accuracy with cost-weighted directional bias penalization substantially reduces absolute physical misallocations (-34.1%) with negligible loss in WMAPE (+0.15 p.p.). By providing fast execution and an intuitive what-If simulation mechanism, the artifact bridges mathematical data science with executive S&OP consensus.

Future work will expand empirical validation across other industrial domains (e.g., heavy machinery, capital goods) and explore dynamic confidence intervals to assist in automated safety stock determination.

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